

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Data Sheets for Isolation Joints
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-009
Document Revision	R01
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Originator	Sunday Ikechukwu
Security Classification	Restricted
Issue Date	27 Jul 2022

Data Sheets for Isolation Joints

R01	27.07.22	Issued for Review	S.Ikechukwu	E.Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	26.07.2022	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-009

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

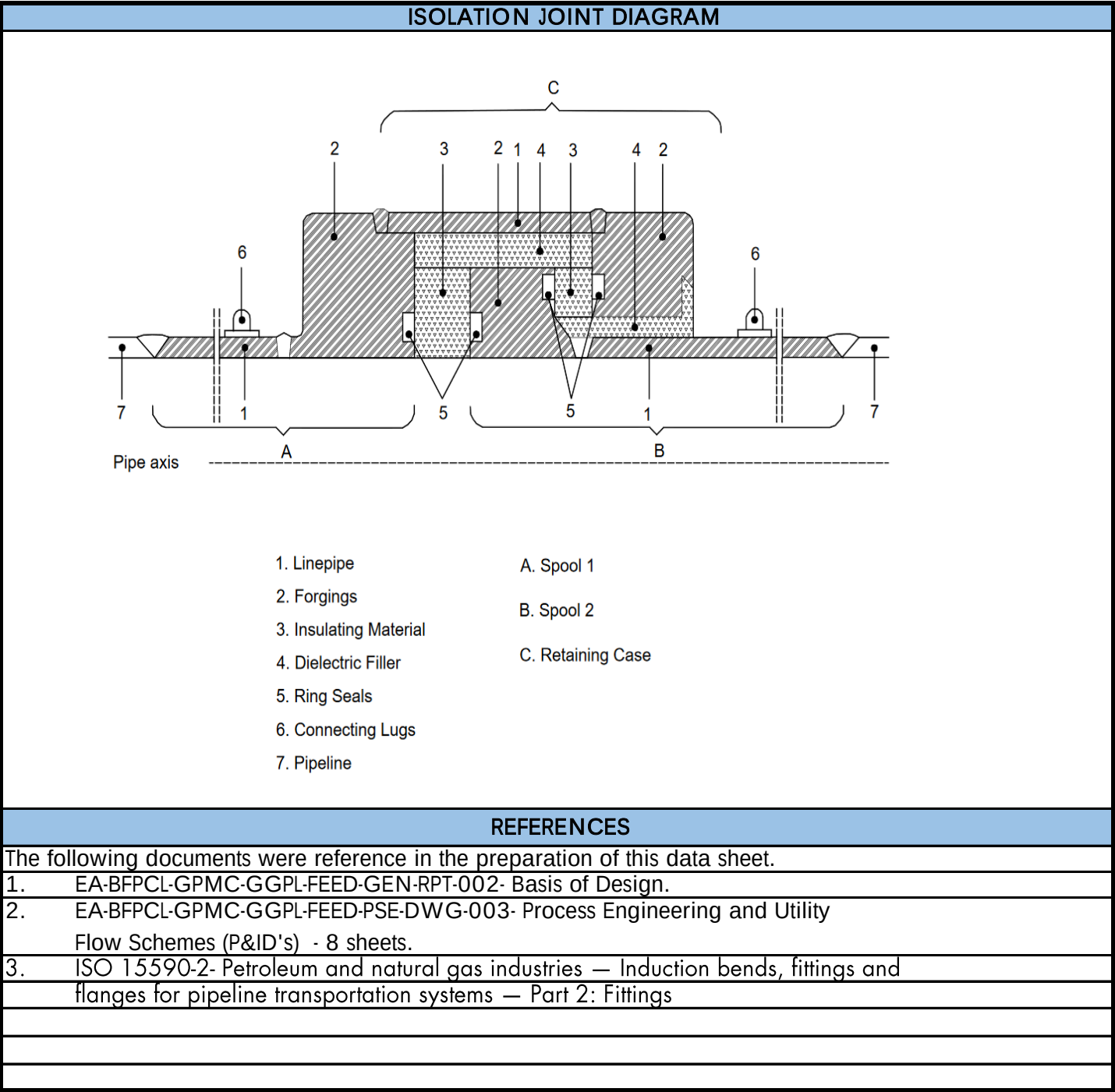
REFERENCES

Document No.	Description

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24" Isolation Joint Data	5
36" Isolation Joint Data	7

ITEM	DESCRIPTION	NOTES
Equipment Name	Monolithic Isolation Joint	
Rating (ASME/ANSI Class)	900#	
Nominal size	24"	
Manufacturer	To be advised	
Piping Special Items Tag No.	To be advised	
Reference P&ID.	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003 Sht 2, sht 5	
Line No.	BBOU - 24"- P - 012 - X70 - CL900	
Location	BBOU/ELEPA Manifold Stations	
Design Code	ASME VIII Div 1	
Transported Fluid	Process Gas	
Quantity	2	
Design Pressure (D.P) (barg)	110	
Design Temperature	-29/60 °C	
Piping Design Code	ASME B31.8	
Corrosion Allowance (mm)	3	
End Connection	Butt-Weld at both ends	1
Hydrostatic Test Pressure (barg)	(1.5*D.P)	2
Charpy Test	Yes: at temp.0°C	3
Materials Specification (Equivalent or Superior)		
Pups	API 5L GR X70	4,9, 6
Body	API 5L GR X70	
Insulating Materials/Elements	See Note 11	
Filler Isolator	Epoxy Resin Cold Cured	
Internal Coating	Amine-Cured Epoxy	
External Coating	Amine-Cured Epoxy	
Connecting Pipe Specification		
Size (inches)	24"	
Wall Thickness(mm)	15.88mm	
Material	API 5L GR X70	
Installation	Above ground	5
Overall length (mm)	Note 12, 6	
NOTES		
1. Butt-welded end are to be prepared in accordance with ASME B16.25		
2. The Supplier/ Vendor shall carry out hydrotest to 1.5*D.P		
3. Charpy test shall be in accordance to ASTM A370.		
4. A length of 50mm at each end shall be left uncoated to allow for welding.		
5. Insulation joints shall be suitable for above ground installation.		
6. Isolating joint shall be inclusive of 500mm pup piece at both ends.		
7. Internal diameter of insulation joints shall not deviate more than -0.4+1.6 mm from the nominal internal diameter of the pipeline.		
8.Welding shall be in accordance with ASME Code IX requirements.		
9. Pipe pups on the ends of the Insulation joints shall be of the same or better as matching line pipe. Material shall be suitable for welding to matching line pipe.		
10. Insulation joints shall have two terminal connecting lugs on the outer surface of each pup ends. Each lug shall be suitable for M10 bolted connection. One lug at each end is intended for electrical measurements. The other lug at each end is intended for a spark gap installation. The lugs shall be welded on additional plates on the pups.		
11: Material shall be proposed by the Manufacturer for Approval by COMPANY.		
12. Figure below is for illustration, manufacturer shall produce a technical datasheet and detailed general arrangement drawing complying to item specification for owners approval.		



ITEM	DESCRIPTION	NOTES
Equipment Name	Monolithic Isolation Joint	
Rating (ASME/ANSI Class)	900#	
Nominal size	36"	
Manufacturer	To be advised	
Piping Special Items Tag No.	To be advised	
Reference P&ID.	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003 Sht 6, sht 8	
Line No.	ELEP- 36"- P - 011 - X70 - CL900 ELEP- 36"- P - 012 - X70 - CL900	
Location	ELEPA Manifold Stations/ Akabel Piging Station	
Design Code	ASME VIII Div 1	
Transported Fluid	Process Gas	
Quantity	2	
Design Pressure (D.P) (barg)	110	
Design Temperature	-29/60 °C	
Piping Design Code	ASME B31.8	
Corrosion Allowance (mm)	6	
End Connection	Butt-Weld at both ends	1
Hydrostatic Test Pressure (barg)	(1.5*D.P)	2
Charpy Test	Yes: at temp.0°C	3
Materials Specification (Equivalent or Superior)		
Pups	API 5L GR X70	4,9, 6
Body	API 5L GR X70	
Insulating Materials/Elements	See Note 11	
Filler Isolator	Epoxy Resin Cold Cured	
Internal Coating	Amine-Cured Epoxy	
External Coating	Amine-Cured Epoxy	
Connecting Pipe Specification		
Size (inches)	36"	
Wall Thickness(mm)	23.83mm	
Material	API 5L GR X70	
Installation	Above ground	5
Overall length (mm)	Note 12, 6	
NOTES		
1. Butt-welded end are to be prepared in accordance with ASME B16.25		
2. The Supplier/ Vendor shall carry out hydrotest to 1.5*D.P		
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